

Optimizing neural network architectures using Genetic Algorithms, Particle Swarm Optimization, and Simulated Annealing

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Overview

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Introduction - Problem Description

- **Given** an image-based dataset with n data points $d_1, d_2, \dots, d_n \in \mathbb{R}^{H \times W \times C}$, where H, W, C represent height, width and channels respectively, and $n \in \mathbb{N}$ represents the number of samples.
- **Find** $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)^\top \in \mathcal{D}^n$ which minimizes

$$f(\alpha, \mathcal{D}) := \lambda_1 \cdot \text{NetworkSize}(\alpha) - \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D})$$

Where:

- α represents the network architecture parameters
- $\mathcal{D}$ is the dataset
- λ_1, λ_2 are weighting factors balancing size vs. accuracy
- $\text{NetworkSize}(\alpha)$ counts the total number of parameters
- $\text{Accuracy}(\alpha, \mathcal{D})$ is the classification accuracy on dataset $\mathcal{D}$

- **Motivation** (Optimization of NNA): Please write :) [1]

Introduction - Architecture Components

Unlike traditional approaches that only focus on fully connected layers, we extend the search space to include the following layers:

- **Convolutional Layers (Conv1D/Conv2D):** Feature extraction with learnable filters
- **Batch Normalization:** Normalization of the values for training stability
- **Max Pooling:** Spatial dimension reduction
- **Dropout:** Regularization technique to avoid overfitting
- **Activation Functions:** ReLU, Tanh, Sigmoid

Introduction - Solution Space Characteristics

- Non-linear cost function
 $f(\alpha, \mathcal{D}) := \lambda_1 \cdot \text{NetworkSize}(\alpha) - \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D})$ combines discrete network size with continuous accuracy metrics
- Mixed discrete-continuous solution space consisting of architecture topology (discrete) and hyperparameters (continuous)
- Solution space can be represented by encoding each architecture α as a variable-length chromosome
- **Naive Approach:** Exhaustive search with runtime $\mathcal{O}(C^L)$ for L max layers with C component types becomes intractable for realistic network sizes

Introduction - Project Objectives

- Investigate how **Genetic Algorithms (GA)**, **Enhanced Genetic Algorithms (EGA)**, **Particle Swarm Optimization (PSO)** and **Simulated Annealing (SA)** perform when applied to the problem.
- Balance the trade-off between network size and accuracy
- Evaluate the effectiveness of different layer types in automated architecture design
- Assess the performance of these algorithms by comparing them to baseline methods, such as **Random Search (RS)** and **Grid Search (GS)**.
- Assess the performance of the algorithms across different dataset complexities

Algorithms - Baseline Methods

Given a image-based dataset $\mathcal{D} = \{d_1, \dots, d_n\}$

Random Search (RS)

Cost function: $f(\alpha^{(i)}) = \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)}) - \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D})$

- 1 Generate N random candidate architectures $\alpha^{(1)}, \dots, \alpha^{(N)}$
- 2 Train each network and evaluate $f(\alpha^{(i)})$ for $i = 1, 2, \dots, N$
- 3 Return $\alpha^{(best, RS)}$ with $f(\alpha^{(best, RS)}) \leq f(\alpha^{(i)}), \forall \alpha^{(i)}$

Grid Search (GS)

Cost function: $f(\alpha^{(i)}) = \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)}) - \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D})$

- 1 Define systematic grid over architecture hyperparameters
- 2 Evaluate $f(\alpha^{(i)})$ for each grid point $\alpha^{(i)}$
- 3 Return $\alpha^{(best, GS)}$ with minimal cost

Algorithms - Genetic Algorithm

Genetic Algorithm (GA) [1]:

Fitness function: $-f(\alpha^{(i)}) = \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)}) - \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D})$

Encoding: Variable-length chromosomes representing layer sequences

- Gene type: {Conv2D, Conv1D, FC, BatchNorm, MaxPool, Dropout}
- Gene parameters: kernel_size, stride, filters, etc.

Algorithm Steps:

- 1 Create `n_pop` random architectures $\alpha^{(i)} = (\text{layer}_1, \text{layer}_2, \dots, \text{layer}_k)$
- 2 Train networks and compute fitness scores
- 3 Tournament selection with `tournament_size` individuals
- 4 Single-point crossover with probability `cxp_b`, preserving valid layer sequences
- 5 Add/remove layers with probability `mut_add` / `mut_del` and modify layer parameters with probability `mut_param`
- 6 Generate new population, repeat for `n_gen` generations
- 7 Return $\alpha^{(\text{best}, (E)GA)}$

Enhanced Genetic Algorithm (EGA) [2]:

Fitness function: $-f(\alpha^{(i)}) = \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)}) - \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D})$

Additional Features:

- Preserve `n_elite` best individuals each generation
- Adjust mutation rates based on population density
- Ensure architectural validity (e.g, Conv2D layers before FC layers)

Algorithms - Particle Swarm Optimization

Particle Swarm Optimization (PSO) [3], [4]:

Position Representation: Continuous vector $x^{(i)} \in \mathbb{R}^d$ encoding the architecture parameters

Velocity Update:

$$v^{(i)}(t+1) = w \cdot v^{(i)}(t) + c_1 \cdot r_1 \cdot (pbest^{(i)} - x^{(i)}(t)) + c_2 \cdot r_2 \cdot (gbest - x^{(i)}(t))$$

Position Update: $x^{(i)}(t+1) = x^{(i)}(t) + v^{(i)}(t+1)$

Architecture Decoding: Map continuous positions to discrete architectures:

- Layer count: $\lfloor x_1 \cdot \text{max_layers} \rfloor$
- Layer types: $\text{argmax}(\text{softmax}([x_2, x_3, \dots, x_7]))$
- Layer parameters: sigmoid/tanh transformations to valid ranges

Algorithm Steps:

- 1 Initialize swarm: Set positions $x^{(i)}(0)$ at random and set velocities $v^{(i)}(0) = 0$
- 2 Convert positions to architectures and train the networks
- 3 Update personal best $pbest^{(i)}$ and global best $gbest$
- 4 Update velocities and positions
- 5 Repeat for `n_iter` iterations
- 6 Return $\alpha^{(best, PSO)}$

Algorithms - Simulated Annealing

Simulated Annealing (SA) [5]:

State Representation: Current architecture $\alpha_{current}$

Neighbourhood Function: Small architectural modifications:

- Add/remove single layer
- Modify layer parameters
- Swap adjacent layers

Algorithm Steps:

- 1 Start with random architecture $\alpha_{current}$
- 2 Apply random modification to generate $\alpha_{neighbour}$
- 3 Acceptance Criterion:
 - If $f(\alpha_{neighbour}) < f(\alpha_{current})$: accept
 - Else: Accept with probability $e^{-\left(\frac{f(\alpha_{neighbour}) - f(\alpha_{current})}{T}\right)}$
- 4 Reduce temperature $T = T_0 \cdot \text{cooling_rate}^{\text{iteration}}$
- 5 Repeat until convergence or `max_iterations`
- 6 Return $\alpha^{(best, SA)}$

Parametric Optimization

All algorithms run on variable parameters:

- Variable parameters for (E)GA:

n_pop	cxp	mutpb	tourn_size	n_gen	n_elite
Population size	Crossover probability	Mutation probability	Tournament size	Generations	Elite size

- Variable parameters for PSO:

n_particles	intw	c ₁	c ₂	n_iter
Swarm size	Inertia weight	Cognitive coefficient	Social coefficient	Iterations

- Variable parameters for SA:

T_0	cooling_rate	max_iter
Initial temperature	Cooling rate	Max. iterations

- Other degrees of freedom include: the type of optimizer used in the neural network, learning rate used in training, type of crossover used in the (E)GA, choice of r_1 and r_2 , etc.
- Fixed parameters used in this work: **HAVE TO ADD THIS**

Simple algorithm to find the best parameters:

```
for each algorithm  $\in \{\text{GA, EGA, PSO, SA}\}$  do  
  for each parameter_set  $\in$  parameter_combinations do  
    for  $k = 1$  to 5 do  
      Split dataset into train/validation  
      Run algorithm with parameter_set  
      Collect: best_fitness, convergence_time, final_accuracy  
    end for  
    Compute: avg_fitness, avg_time, avg_accuracy  
  end for  
  Select parameter_set with best avg_fitness  
end for
```

Dataset Splits:

- Training: 60% (Architecture search)
- Validation: 20% (Fitness evaluation)
- Test: 20% (Final performance assessment)

Parametric Optimization - Genetic Algorithm

Parametric Optimization - Particle Swarm Optimization

Parametric Optimization - Simulated Annealing

Results

Future Work

Conclusion and Discussion

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